

Sample Questions for Class Test I CITS2200 Data Structures and Algorithms

Q1. The time complexity of the Insertion Sort algorithm is best expressed as (n is the size of the input)

(a) $O(n \log n)$; (b) $O(\log n)$; (c) $O(n^3)$; (d) $O(n^4)$. c

Q2. The time complexity of the Merge Sort algorithm is best expressed as:

(a) $O(n^2)$; (b) $O(n)$; (c) $O(n \log n)$; (d) none of the above. c

Q3. The following is Python code for binary search with a comparison in the second line missing (indicated as '?'). The condition should be:

```
def binary_search(data, target, low, high)
    if ?:
        return False
    else:
        mid = (low+high)//2
        if target == data[mid]
            return True
        elif target < data[mid]:
            return binary_search(data, target, low, mid-1)
        else:
            return binary_search(data, target, mid+1, high)
```

d

(a) $low > mid$; (b) $low \neq high$; (c) $low < high$; (d) $low > high$

Q4. Which of the following statements about the worst-case complexity of Insertion Sort is wrong: a

(a) Worst-case complexity of Insertion Sort is $O(n)$; (b) Worst-case complexity of Insertion Sort is $O(n^2)$; (c) Worst-case complexity of Insertion Sort is $O(n^{20})$; (d) Worst-case complexity of Insertion Sort is $O(n^3)$.

Q5. Suppose $f(n)$ is $O(g(n))$, $g(n)$ is $O(h(n))$, and $h(n)$ is $O(f(n))$. Which of the following are possible functions for f , g and h ?

(a) $f(n) = \log^2 n$, $g(n) = n \log n$, $h(n) = n^2$; (b) $f(n) = n^3$, $g(n) = n^2$, $h(n) = n \log n$; (c) $f(n) = 5 \log n$, $g(n) = 1000 \log n$, $h(n) = \log n$; (d) $f(n) = n^2$, $g(n) = n^4$, $h(n) = 2^n$ c

Q6. Suppose $f(n) = 2^{\log n}$, $g(n) = n^2$, $h(n) = n\sqrt{n}$, $k(n) = n \log n$, $p(n) = 2^n$. Which of the following is a correct ordering of these complexities in ascending order (smallest to largest)?

(a) $f(n), h(n), g(n), p(n), k(n)$; (b) $f(n), k(n)h(n), g(n), p(n)$; (c) $h(n), k(n), f(n), g(n), p(n)$; (d) $p(n), k(n), f(n), g(n), h(n)$ b

Q7. The correct recurrence equation for analysing the complexity of the Merge Sort algorithm is (c is a constant):

(a) $T(n) = T(\frac{n}{2}) + cn$; (b) $T(n) = T(n-1) + cn$; (c) $T(n) = 4T(\frac{n}{2}) + cn$; (d) $T(n) = 2T(\frac{n}{2}) + cn$ d

Q8. There are three parts in an algorithm with complexities $n \log n$, $\log n$ and n^2 . What is the overall complexity of the algorithm? d

(a) $O(n \log n)$; (b) $O(n^3 \log n)$; (c) $O(n^3 \log^2 n)$; (d) $O(n^2)$.

Q9. You are given two sorted arrays A and B of size m and n . You have to check whether each element of array A is in array B . The best time complexity for an algorithm for this task is: c

(a) $O(n \log n)$; (b) $O(mn)$; (c) $O(m \log n)$; (d) $O(m^2)$.

Q10. Given an unsorted array of n numbers, what is the complexity of the best algorithm for finding the minimum number in the array? a

(a) $O(n)$; (b) $O(n \log n)$; (c) $O(n^2)$; (d) $O(n^2 \log n)$.